Fiscal Year:	FY 2007	**Task Last Updated:**	FY 08/10/2007
PI Name:	Globus, Ruth Ph.D.		
Project Title:	Simulated Microgravity and Radiation-Induced Bone Degeneration: Oxidative Stress- and p53-Dependent Mechanisms		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes (3) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Ruth.K.Globus@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650-604-5247
Organization Name:	NASA Ames Research Center		
PI Address 1:	Bone and Signaling Laboratory		
PI Address 2:	Space Biosciences Research Branch		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2004 Radiation Biology NNH04ZUU005N
Start Date:	10/01/2005	**End Date:**	09/30/2009
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes at that time, per JSC/A. Chu-ARC (jvp 4/2009)		
Key Personnel Changes/Previous PI:	none		
COI Name (Institution):	Almeida, Eduardo A.C. (NASA Ames Research Center) Limoli, Charles (University of California Irvine) Searby, Nancy (NASA Ames Research Center) Vercoutere, Wenonah (NASA Ames Research Center)		
Grant/Contract No.:	INTRAMURAL		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	To determine the risks and mechanisms of tissue degeneration in the spaceflight environment, we will examine the combined effects of radiation and simulated weightlessness on bone. We will focus our studies on two important pathways, p53 and oxidative and nitrative stress-related species, in bone marrow progenitor populations because there is abundant evidence that the biological effects of weightlessness and those of radiation share these mechanism. Experiments will be conducted using the rodent hindlimb unloading model, the best validated animal model for simulating weightlessness, followed by irradiation. In Aim 1, we will test the hypothesis that simulated weightlessness of mice modulates the response of bone and bone marrow to radiation, both acutely and during tissue regeneration. In Aim 2 we will identify the target marrow-derived stem, progenitor and differentiated cell populations sensitive to the combined stimuli of simulated weightlessness and radiation. In Aim 3, we will explore the relevance of specific ROS/RNS and p53-dependent mechanisms for mediating cellular responses to combined simulated weightlessness and radiation. The results of these studies are expected to provide important information for the development of countermeasures both to help prevent degeneration of connective tissues such as bone during spaceflight and to improve subsequent tissue regeneration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project addresses several of the risks identified in NASA's Bioastronautics Critical Path Roadmap (BCPR) caused by long duration spaceflight. We will obtain basic new knowledge about bone tissue degeneration caused by musculoskeletal disuse and radiation at the molecular, cell, and tissue level, relevant to Earth, as well as in space. Results will also lead to the formulation of new countermeasures concepts and risk assessment strategies for bone and connective tissue degenerative diseases.
Task Progress:	Our initial studies have demonstrated conclusively that both irradiation and hypergravity induce increased oxidative stress in the form of elevated reactive oxygen species (ROS) when applied to cells cultured over 2-5 days after isolation. Interestingly, the increased ROS levels derived from either stress translated into somewhat divergent physiological effects. While radiation treatments led to the expected cell kill, hypergravity likely increased cellular proliferation and/or adhesion that increased the number of cells capable of anchoring to the substrate. Cells subjected to irradiation were inhibited in their ability to differentiate, while differentiation was actually promoted in cells subjected to hypergravity. Thus, while irradiation and hypergravity elicit similar changes in cellular redox state, the physiological consequences are quite different. These data suggest that multiple redox sensitive pathways interact to elicit physiologic change in bone marrow stromal cells. Our recent results showed that irradiation alone decreased cancellous bone mass, likely due to an increase in bone resorption. Oxidative damage is increased in bone by gamma irradiation, suggesting the generation of reactive oxygen species following irradiation stimulates osteoclastogenesis. We conclude irradiation caused oxidative stress in bone tissue that may stimulate the recruitment and activation of osteoclasts to bone surfaces, leading to a rapid loss of cancellous bone. Both hindlimb unloading (to simulate weightlessness) and irradiation cause bone loss, but possibly via different molecular mechanisms. These results have potential clinical relevance to astronauts as well as to cancer patients treated therapeutically with radiation.
Bibliography Type:	Description: (Last Updated: 06/04/2025)
Abstracts for Journals and Proceedings	Kondo H, Phillips J, Limoli C, Almeida EAC, Loftus DJ, Vercoutere W, Morey-Holton E, Mojarrab R, Globus RK, Searby ND. "A single dose of gamma or proton radiation rapidly compromises skeletal structure of adult mice." International Conference for Radiation Research, San Francisco, CA, July 2007. *Note that Dr. Kondo received a student travel Award for his presentation at this meeting. International Conference for Radiation Research, San Francisco, CA, July 2007. , Jul-2007
Abstracts for Journals and Proceedings	Kondo H, Mojarrab R, Wang A, Phillips J, Almeida EAC, Loftus DJ, Morey-Holton E, Globus RK, Vercoutere W, Limoli C, Searby ND. "Cellular mechanisms for acute cancellous bone loss in adults caused by radiation and simulated weightlessness." 18th Annual NASA Space Radiation Health Investigators Workshop, Rohnert Park, CA, July 2007. 18th Annual NASA Space Radiation Health Investigators Workshop, Rohnert Park, CA, July 2007. , Jul-2007
Abstracts for Journals and Proceedings	Kondo H, Mojarrab R, Wang A, Phillips J, Almeida EAC, Loftus DJ, Morey-Holton E, Globus RK, Vercoutere W, Limoli C, Searby ND. "Increased active osteoclasts are associated with acute cancellous bone loss in adult mice exposed to ionizing radiation and musculoskeletal disuse." To be presented at American Society for Bone and Mineral Research annual meeting, Honolulu HI, September 2007. American Society for Bone and Mineral Research annual meeting, Honolulu HI, September 2007. , Sep-2007
Abstracts for Journals and Proceedings	Kondo H, Limoli C, Searby ND, Almeida EAC, Loftus DJ, Vercoutere W, Hilton D, Globus RK. "Shared Oxidative Pathway in Response to Gravity-Dependent Loading and Gamma Irradiation of Bone Marrow-Derived Skeletal Cell Progenitors." Presented at the 17th Annual NASA Space Radiation Health Investigators Workshop, Moscow-St. Petersburg, Russia, June 2006. 17th Annual NASA Space Radiation Health Investigators Workshop, Moscow-St. Petersburg, Russia, June 2006. , Jun-2006
Articles in Peer-reviewed Journals	Kondo H, Limoli C, Searby ND, Almeida EAC, Loftus DJ, Vercoutere W, Morey-Holton E, Giedzinski E, Mojarrab R, Hilton D, Globus RK. "Shared oxidative pathways in response to gravity-dependent loading and gamma irradiation of bone marrow-derived skeletal cell progenitors. " Radiation Biology and Radioecology. 2007;47(3):281-5. , Jan-2007
Awards	Kondo H, Mojarrab R, Wang A, Phillips J, Almeida EAC, Loftus DJ, Morey-Holton E, Globus RK, Vercoutere W, Limoli C, Searby ND. "Student poster award for Dr. H. Kondo's presentation: Cellular mechanisms for acute cancellous bone loss in adults caused by radiation and simulated weightlessness, 18th Annual NASA Space Radiation Health Investigators Workshop, Rohnert Park, CA, July 2007. " Jul-2007